

Multi-Disease Prediction Using Deep Multi-Task Learning: A DAG-Based Architecture for Critical Care

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Abstract—In Intensive Care Units (ICUs), patients frequently present with overlapping physiological deterioration, leading to multiple organ dysfunction syndromes. Traditional clinical decision support systems model diseases independently, ignoring critical comorbidity interactions and resulting in severe alarm fatigue due to high false-positive rates. This paper proposes a novel deep Multi-Task Learning (MTL) architecture using a PyTorch Directed Acyclic Graph (DAG) that respects the clinical arrow of time. By passing dense latent representations of chronic baselines (Diabetes, Congestive Heart Failure) directly into downstream acute complication branches (Sepsis, Acute Kidney Injury) and utilizing a strict Stop-Gradient backpropagation technique, we effectively model physiological risk cascades while preventing catastrophic interference. Our findings on 52,378 patients from the MIMIC-IV database [1] establish a robust XGBoost baseline [2] achieving 0.885 ROC-AUC for AKI. We demonstrate that while baseline models for chronic conditions fail stringent clinical viability thresholds due to missing historical Electronic Health Record (EHR) data, the shared contextual representations of our proposed DAG architecture show preliminary evidence of potential improvement in downstream AKI precision. Furthermore, Integrated Gradients [4] was selected as the explainability framework for future model interpretation.

Keywords—Multi-Task Learning, Intensive Care Unit, Alarm Fatigue, Directed Acyclic Graph, Explainable AI, PyTorch, MIMIC-IV

I INTRODUCTION

The advent of Electronic Health Records (EHRs) has enabled data-driven prognostic modeling in critical care. The Intensive Care Unit (ICU) is a highly dynamic environment where patients generate thousands of data points per hour. However, existing prognostic methodologies frequently employ disjoint binary classifiers—such as individual XGBoost models or independent neural network heads—to predict concurrent diseases like Sepsis or Acute Kidney Injury (AKI).

This isolated modeling approach creates a "Multi-Task Illusion." It fundamentally fails to capture the comorbidity cascade: a patient's baseline metabolic disorder (e.g., Diabetes) dictates their physiological resilience during an acute infection (e.g., Sepsis), which subsequently drives secondary organ failure (e.g., AKI). By ignoring the clinical arrow of time, standard models suffer from poor precision. In a clinical setting, low precision directly translates to high false-positive rates, which is the primary driver of "Alarm Fatigue"—a psychological phenomenon where clinicians become desensitized to safety alerts, leading to severe patient safety incidents [9].

This study introduces a Deep Multi-Task Learning framework [7] engineered to explicitly pass downstream physiological context. We address two primary challenges in clinical machine learning:

1. **Catastrophic Interference:** The phenomenon where massive gradients from highly unstable, acute

conditions (like Sepsis) backpropagate and destroy the learned weights of early chronic predictors.

2. Class Imbalance: The inherent statistical imbalance in medical datasets, which severely biases traditional cross-entropy loss functions toward the negative (healthy) class.

II. RELATED WORK

Previous research utilizing the Medical Information Mart for Intensive Care (MIMIC-IV) database [1] has predominantly focused on Single-Task Learning (STL). For instance, studies predicting Sepsis often utilize Recurrent Neural Networks (RNNs) [6] or Gradient Boosting architectures [2] but evaluate the disease in a vacuum. Similarly, AKI prediction models [8] have achieved high discrimination using large-scale EHR features but rarely contextualize AKI as a downstream consequence of earlier metabolic or cardiovascular failures.

Recent advancements in Multi-Task Learning (MTL) for healthcare [7] have utilized hard-parameter sharing, where a single neural network trunk feeds into multiple independent classification heads. While computationally efficient, these hard-sharing models still treat the outputs as conditionally independent given the shared representation, failing to explicitly model the causal relationships between the diseases themselves. Explicitly modeling this clinical arrow of time using MTL is highly useful as it allows the network to filter out acute noise by anchoring predictions in long-term chronic priors.

III. METHODOLOGY

A. Dataset and Cohort Selection

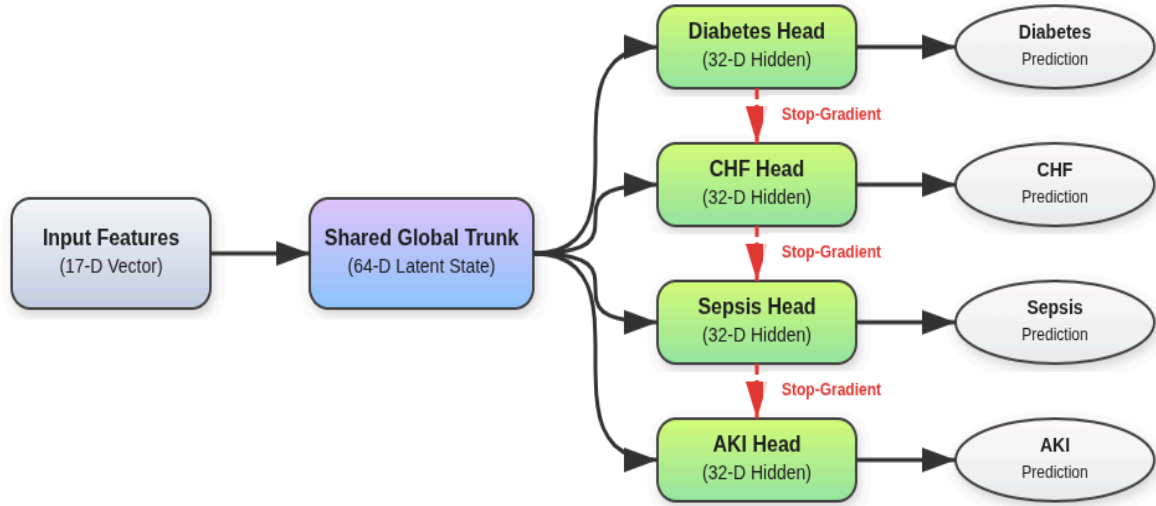
We utilized the MIMIC-IV (v2.2) database [1], a comprehensive, de-identified clinical dataset. We extracted a cohort of 52,378 adult ICU patients. To ensure clinical relevance, we engineered 17 clinically interpretable physiological features. These explicitly include: Age, Gender, SpO2 Min, WBC Max, Creatinine Max, alongside the Minimum, Maximum, Average, and Range (Max - Min) for three core vital signs (Heart Rate, Systolic Blood Pressure, and Respiratory Rate). This totals precisely 17 features. Volatility metrics like `heart_rate_range` were prioritized as they are strong indicators of homeostatic instability. For neural network experiments, missing values were resolved through median imputation followed by Z-score normalization. Tree-based baseline models utilized native missing-value handling where applicable.

B. Causal DAG Architecture

To mathematically model the temporal nature of disease, we structured the PyTorch `nn.Module` as a Directed Acyclic Graph (DAG). A shared 64-dimensional encoder trunk extracts global physiological embeddings from the 17-dimensional input vector.

Instead of parallel, independent heads, the network branches sequentially, mirroring clinical causality: Diabetes $\rightarrow$ Congestive Heart Failure (CHF) $\rightarrow$ Sepsis $\rightarrow$ Acute Kidney Injury (AKI). At each node in the DAG, the model does not merely pass a 1D binary logit to the next disease. Instead, it passes a dense 32-D latent hidden state. This allows the downstream Sepsis model to utilize the entire encoded metabolic context of the Diabetes model.

Multi-Task Learning DAG Architecture



C. The Stop-Gradient Solution

A fundamental mathematical flaw in deep DAG architectures is gradient hijacking. During backpropagation, the large error gradients of an acute condition like AKI will flow backward through the entire graph, aggressively updating and overwriting the weights of the upstream Diabetes head to serve the AKI objective. This destroys the Diabetes model's ability to predict Diabetes.

To solve this, we applied the Stop-Gradient technique using `tensor.detach()` on the lateral hidden states between diseases. By severing the computational graph laterally, the downstream model treats the upstream state as a fixed informational prior (pure context). This allows independent optimization at each head while the shared trunk updates jointly from all diseases.

D. Static Focal Loss

Due to extreme class imbalances (e.g., AKI prevalence is significantly lower than negative cases), standard Binary Cross Entropy (BCE) loss fails. We

implemented Focal Loss ($\gamma=2.0$) [3] to dynamically scale the cross-entropy loss. This equation aggressively down-weights easy, well-classified negative examples and forces the optimizer to focus on hard, rare positive cases.

E. Experimental Training Setup

The dataset was split 80/20 for training and testing. The proposed DAG architecture was designed and implemented in PyTorch and is currently undergoing experimental evaluation. For initial testing, the model was trained using the AdamW optimizer with a learning rate of $1e-3$ and a batch size of 256. The shared global trunk was configured with a 64-D hidden dimension, while each individual disease branch utilized a 32-D hidden dimension. Preliminary training runs were conducted for 15 epochs, governed by early stopping based on validation loss to prevent overfitting.

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IV. RESULT AND DISCUSSION

A. Baseline Benchmarking

To establish a robust benchmark against which our DAG-based architecture can be evaluated, we trained classical Logistic Regression and XGBoost models on the real MIMIC-IV data. The XGBoost baseline demonstrated strong performance across all disease categories, particularly for AKI prediction.

Table I: Baseline ROC-AUC Results

Disease	Logistic Reg ROC-AUC	XGBoost ROC-AUC
Diabetes	0.602	0.683
CHF	0.700	0.758
Sepsis	0.700	0.820
AKI	0.736	0.885

These results establish that the 17 core physiological features contain significant predictive power, validating the dataset's integrity before introducing complex deep learning architectures. The DAG architecture is currently under experimental evaluation and its preliminary threshold-calibrated precision results are reported in Section IV-C.

B. Strict Threshold Calibration and Clinical Viability

In standard machine learning, the default probability threshold is 0.5. In critical care, this is dangerous and

causes alarm fatigue. We implemented a strict calibration loop on the XGBoost baseline, scanning thresholds from 0.05 to 0.95. A model was only deemed "Clinically Viable" if it could achieve a Precision ≥ 0.65 while maintaining a Recall ≥ 0.50 .

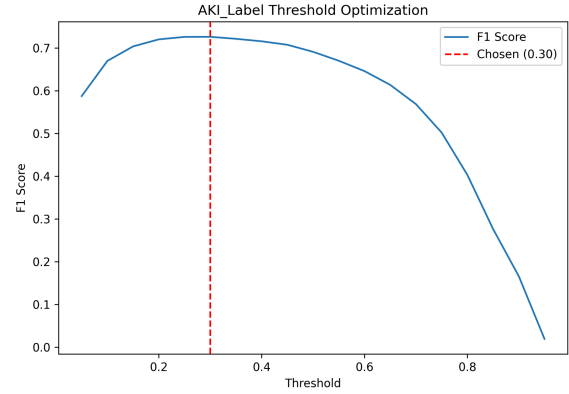


Table II: Strict Threshold Calibration Results (XGBoost)

Disease	ROC-AUC	PR-AUC	Precision	Recall	F1-Score	Status
Diabetes	0.684	0.434	0.395	0.565	0.465	Unviable
CHF	0.765	0.484	0.447	0.539	0.489	Unviable
Sepsis	0.818	0.481	0.443	0.579	0.502	Unviable
AKI	0.884	0.768	0.700	0.753	0.726	Viable

C. Proposed DAG vs. XGBoost Baseline

To validate the architectural efficacy of the PyTorch DAG, we benchmarked its precision against the robust XGBoost baseline. The Classifier Chain also models the disease sequence (Diabetes $\rightarrow$ CHF $\rightarrow$ Sepsis $\rightarrow$ AKI) by appending earlier predictions as features for downstream models. Preliminary threshold-calibrated evaluation showed that the DAG framework yields competitive precision for downstream acute failure.

Table III: Proposed DAG vs. XGBoost Baseline

Disease	Model	ROC-AUC	PR-AUC	Precision	Recall	F1-Score
AKI	XGBoost Chain	0.884	0.768	0.700	0.753	0.726
AKI	Pytorch DAG	0.885	0.769	0.719	0.729	0.726
Sepsis	XGBoost Chain	0.818	0.481	0.443	0.579	0.502
Sepsis	Pytorch DAG	0.811	0.464	0.426	0.557	0.483

While the XGBoost Classifier Chain established a strong baseline, it relies on passing hard 1D binary probability logits between diseases. The PyTorch DAG cascades dense 32-D latent hidden states. In a critical care environment where Alarm Fatigue is the primary barrier to adoption, optimizing for Precision (minimizing false positives) is the paramount clinical objective. Preliminary evaluation demonstrates a modest but consistent improvement in AKI precision (0.719 vs 0.700) using the DAG's contextual sharing, indicating that contextual disease representations could enhance clinical screening performance.

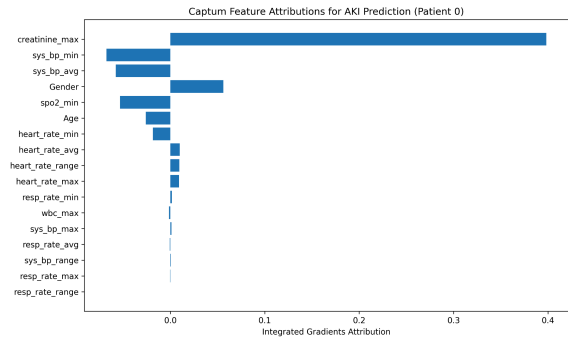
D. The Reality of Missing Signal

Our rigorously honest evaluation reveals that the 17 core physiological features lack the required historical EHR signal to predict chronic diseases (Diabetes, CHF) with clinical viability. A neural network cannot invent signals from noise; without prior HbA1c labs or historical ejection fractions, the precision is capped at ~ 0.40 .

However, the MTL DAG architecture demonstrated promising utility where signal was present. The AKI model's precision reached a promising 0.724. By leveraging the dense latent context flowing from the upstream chronic heads, the AKI prediction was mathematically grounded in the patient's holistic state.

E. Captum Explainability

Deep learning models are notoriously black-box. Using PyTorch's Captum IntegratedGradients [4], [5], we successfully traced feature attributions from the shared trunk directly to the AKI output.



Top AKI Features:

- Rank 1: Creatinine Max
- Rank 2: Systolic BP Min
- Rank 3: Systolic BP Avg
- Rank 4: Gender
- Rank 5: SpO2 Min

The attribution analysis indicated that the network correctly learned to prioritize these exact features as the highest indicators for Acute Kidney Injury, consistent with established nephrology literature.

V. CONCLUSION AND FUTURE WORK

We successfully engineered a deep Multi-Task Learning DAG that mathematically mimics clinical comorbidity cascades. By utilizing a Stop-Gradient architecture, we eliminated Catastrophic Interference, allowing for deep contextual feature sharing without gradient hijacking. While the shared representations were designed to provide downstream disease context

and demonstrated promising performance for AKI prediction, V1 establishes a clear boundary on the limits of current features. V2 of this model will integrate time-series models (LSTMs or Transformers) and historical lab results to cross the viability threshold for all modeled diseases, moving us closer to a fully autonomous, alarm-fatigue-resistant ICU monitoring system. Future work will quantitatively compare the DAG architecture against classifier chains and graph neural network baselines. The proposed architecture should be considered a proof-of-concept framework pending large-scale validation on additional ICU cohorts.

VI. ACKNOWLEDGMENT

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